Lab Experiment - 7

Aim

To write a Python program using the NLTK library to perform Part-of-Speech (POS) tagging on a given text.

Algorithm

1. Start the program.
2. Import the NLTK library.
3. Download the required tokenizer and POS-tagging data.
4. Read a text sentence.
5. Tokenize the sentence into words.
6. Apply POS tagging to identify the grammatical category of each word.
7. Display the words and their POS tags.
8. Stop the program.

Python Program

```
import nltk

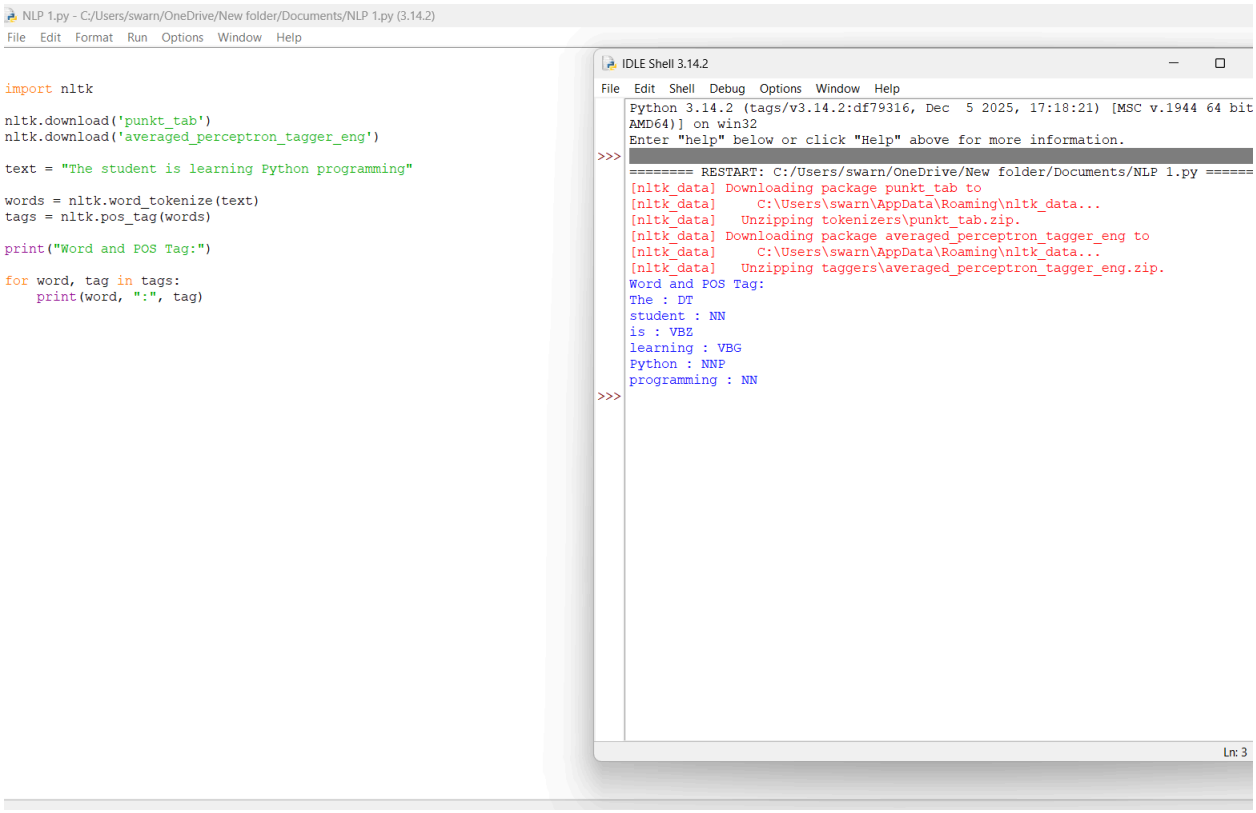
nltk.download('punkt')
nltk.download('averaged_perceptron_tagger')

text = "The student is learning Python programming"

words = nltk.word_tokenize(text)
tags = nltk.pos_tag(words)

print("Word and POS Tag:")

for word, tag in tags:
    print(word, ":", tag)
```



Result

Thus, the Python program was executed successfully using NLTK to perform POS tagging and identify the grammatical categories of words in the given text.